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A Framework for Rigorous Evaluation of Model-Agnostic Explainability Methods: Multi-Metric Statistical Benchmarking, Operational Protocol, and Reproducibility

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Abstract

Evaluating explainability methods requires more than a single faithfulness proxy. We present a modular benchmarking framework centered on quantitative XAI quality metrics: fidelity, stability, sparsity, computational cost, and faithfulness gap, plus an explicit method for operating the framework end-to-end. On the UCI Adult benchmark, we use a staged protocol (EXP1 calibration/reproducibility and EXP2 comparative/robustness benchmarking); the robustness cohort currently contains 300 of 300 planned configurations (100.0% coverage). Across complete model-size blocks (5 models, $N \in \{50, 100, 200\}$), Friedman tests indicate significant method differences for fidelity ($\chi^2 = 42.12$, $p = 3.78 \times 10^{-9}$), stability ($\chi^2 = 43.88$, $p = 1.60 \times 10^{-9}$), sparsity ($\chi^2 = 35.64$, $p = 8.92 \times 10^{-8}$), faithfulness gap ($\chi^2 = 45.00$, $p = 9.25 \times 10^{-10}$), and runtime ($\chi^2 = 27.72$, $p = 4.16 \times 10^{-6}$). SHAP leads on fidelity/stability, DiCE leads on sparsity, and LIME is generally fastest and most practical outside SVM-KernelSHAP bottlenecks. We release the framework, operation protocol, and artifacts with explicit data-quality caveats for reproducible benchmark use under a quantitative-only claim scope.

Keywords: Explainable AI, benchmark methodology, reproducibility, statistical evaluation, model-agnostic explanations

1 Introduction

1.1 Problem Setting and Gap

Evaluation of post-hoc explainability methods remains methodologically fragmented. Recent reviews show heterogeneous metric choices and inconsistent evaluation objectives across domains, which weakens comparability and cumulative progress (Kadir et al., 2023; Pawlicki et al., 2024; Adadi and Berrada, 2018; Arrieta et al., 2020). Fidelity-centric reporting is still common even when stability, complexity, and computational burden materially affect practical usefulness, especially in high-stakes settings (Rudin, 2019).

Tooling has improved implementation-level reproducibility, including interfaces for broad explainer and metric coverage (Li et al., 2022; Hedström et al., 2023). However, tool availability does not establish inferential comparability by itself. Benchmark claims still require explicit controls for split discipline, artifact qualification, analysis units, and multiplicity-aware statistical inference. Canonical model-agnostic explanation families (local surrogates, additive attributions, rule-based anchors, and counterfactual recourse) also rely on different

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assumptions, which complicates fair cross-method comparison (Ribeiro et al., 2016; Lundberg and Lee, 2017; Ribeiro et al., 2018; Mothilal et al., 2020).

1.2 Related Work and Novelty Delta

Recent benchmark efforts in graph explainability show that value depends on explicit dataset assumptions, ground-truth design, and transparent reporting constraints (Agarwal et al., 2023; Proszewska et al., 2025). Related findings on attribution consensus in Rashomon sets further indicate that explanation behavior can vary even under comparable predictive performance (Laberge et al., 2023). Faithfulness-oriented work also shows sensitivity to perturbation and protocol choices (Zheng et al., 2025).

The present study addresses a gap left by existing tools and benchmarks: a single, auditable protocol that links benchmark execution governance, artifact quality control, and inferential reporting. The methodological contribution is not a new explainer algorithm; it is a reproducible benchmark operation model that connects design decisions to claim eligibility.

1.3 Contributions and Scope

This paper contributes:

1. a model-agnostic benchmark framework spanning five model families and four explainer families under shared execution controls;
2. a staged evaluation protocol (EXP1/EXP2) with explicit artifact qualification and claim-role separation;
3. a multi-metric inferential stack combining Friedman omnibus tests, Nemenyi localization, and paired Wilcoxon contrasts with multiplicity control;
4. an auditable **F**ramework **O**peration **M**ethod with seven gates (FOM-7) for protocol freeze, controlled execution, integrity audit, inferential export, and claim-traceable reporting.

FOM-7 emerged while consolidating early pilot/calibration cycles (EXP1), where protocol drift, malformed artifacts, and ad-hoc analysis-table harmonization repeatedly threatened claim comparability. The method was therefore formalized as a seven-gate operation protocol that makes claim eligibility auditable.

Scope is intentionally bounded to model-agnostic tabular post-hoc explanation benchmarking on Adult Income with quantitative metrics (fidelity, stability, sparsity, faithfulness gap, and cost). The remainder of this paper presents the methodology (Section 2), reports comparative and reproducibility evidence (Section 3), and concludes with implications, limitations, and next steps (Section 4).

2 Methodology

2.1 Study Scope and Evidence Cohorts

Benchmarking literature identifies two recurring barriers to reliable XAI comparison: non-standardized evaluation constructs and weak traceability between execution artifacts and reported claims. Recent frameworks and toolkits emphasize explicit property definitions,

Table 1: Evidence cohorts and their role in this study's claims.

Cohort	Primary objective	Role in claims
EXP1 pilot/calibration	Implementation sanity checks before scaled runs.	Not used for confirmatory significance claims.
EXP1 reproducibility	Repeated-seed variance profiling on core settings.	Supports reproducibility characterization (dispersion/CV).
EXP2 comparative	Broad method-model coverage under fixed seeds.	Descriptive comparative context.
EXP2 robustness	Multi-seed, multi-sample-size benchmark runs.	Primary inferential cohort for global and paired tests.

transparent benchmarking pipelines, and artifact-level reproducibility (Canha et al., 2025; Agarwal et al., 2022; Sithakoul et al., 2024; Hedström et al., 2023; Adadi and Berrada, 2018; Arrieta et al., 2020).

Following this guidance, this study separates evidence generation into staged cohorts with explicit claim roles. The separation is analogous to train/validation/test discipline, but applied to benchmark evidence rather than predictive model fitting. This design reduces leakage between calibration activities and confirmatory inference, a risk discussed in recent faithfulness evaluation work (Zheng et al., 2025), and it makes missingness/exclusion decisions auditable before inferential claims are finalized.

2.2 Research Questions, Factors, and Analysis Units

The confirmatory questions are:

1. **RQ1 (global separation):** do explainer methods differ on the primary metrics under matched contexts?

2. **RQ2 (targeted pairwise contrast):** on matched cells, does SHAP differ from LIME on quality-cost trade-offs?

3. **RQ3 (reproducibility profile):** how much metric dispersion appears under repeated runs?

EXP2 is a crossed design with model family g, explainer method k, seed s, and per-quadrant sampling intensity n:

$$g \in \{\text{logreg}, \text{rf}, \text{xgb}, \text{svm}, \text{mlp}\}, \quad k \in \{\text{shap}, \text{lime}, \text{anchors}, \text{dice}\},$$

$$s \in \{42, 123, 456, 789, 999\}, \quad n \in \{50, 100, 200\}.$$

Planned EXP2 robustness size is $5 \times 4 \times 5 \times 3 = 300$ runs. In the current snapshot, 300 artifacts are present and 283 are analyzable (16 empty, 1 malformed).

The inferential unit is not the instance row. For metric m, the run-level estimand is $y_{g,k,s,n,m}$ and the matched block is $b = (g, n)$. Tests are performed on run/block aggregates to avoid pseudo-replication.

Table 2: EXP2 evidence accounting for inferential qualification.

Qualification stage	**Count**	**Share of planned runs**
Planned robustness grid ($5 \times 4 \times 5 \times 3$)	300	100.0%
Present artifacts	300	100.0%
Analyzable artifacts	283	94.3%
Excluded artifacts (16 empty + 1 malformed)	17	5.7%
Complete (g, n) blocks used by Friedman tests	15/15	100.0% block completion

Planned 300 → **Present** 300 → **Analyzable** 283 → **Friedman-ready blocks** 15/15
Exclusions before inference: 16 empty artifacts + 1 malformed artifact.

Figure 1: Evidence qualification waterfall for EXP2 robustness inference.

2.3 Data Protocol and Leakage Controls

The benchmark uses Adult Income loaded through `src/data_loading/adult.py` with OpenML as primary source and UCI fallback. Cleaning applies whitespace normalization, missing-value canonicalization, target normalization, duplicate removal, and deterministic feature ordering. Deterministic preprocessing and artifact traceability follow reproducibility recommendations in open benchmarking toolkits (Agarwal et al., 2022; Hedström et al., 2023).

Leakage controls are fixed across all explainers:

1. stratified split before any preprocessing fit,
2. train-only preprocessor fitting,
3. shared persisted preprocessor across methods in each run,
4. explanation on the same transformed feature space used by the model.

Data split is

```
train_test_split(test_size=0.2, stratify=y, random_state=seed).
```

For seed 42, the split is 39,032 train and 9,758 test rows. Each run samples TP/TN/FP/FN strata with target N per stratum (nominal run size $4N$, subject to stratum capacity).

2.4 Model and Explanation Protocol

EXP2 evaluates frozen EXP1 model artifacts; models are not retrained during EXP2. Model families are `logreg`, `rf`, `xgb`, `svm`, and `mlp`, with fixed repository-tracked settings.

All methods are executed through `ExplainerWrapper.explain_instance` under a standardized runtime constraint protocol (Cores=1, RAM=4GB, Timeout=300s) enforced by the infrastructure's resource guard to ensure comparable wall-clock duration reporting. Effective explainer conditions are:

Table 3: Primary metric orientation for objective-consistent interpretation.

Metric	Unit	Direction	Operational interpretation
Fidelity	score	↑	Larger perturbation agreement indicates stronger local faithfulness under top-k masking.
Stability	cosine sim.	↑	Higher perturbation consistency indicates less explanation volatility.
Sparsity	score	↓	Lower value indicates more concise explanations under this implementation.
Faithfulness Gap	score	↑	Larger sensitivity gap indicates stronger alignment between importance ranking and predictive effect.
Cost	ms	↓	Lower runtime indicates higher practical deployability.

1. SHAP: tree mode for RF and XGB; kernel mode for LogReg, SVM, and MLP; background sample count set to 50 (Lundberg and Lee, 2017).
2. LIME: `num_samples=1000`, `num_features=10`, `kernel_width=3.0` (Ribeiro et al., 2016).
3. Anchors: `AnchorTabular` on transformed training data with effective threshold fixed at 0.95 (Ribeiro et al., 2018).
4. DiCE: random backend, opposite-class counterfactual target, importance from absolute original–counterfactual deltas (Mothilal et al., 2020).

Two implementation-bound caveats are explicit in claim interpretation: Anchors YAML threshold is not consumed at runtime (fixed 0.95), and DiCE YAML counterfactual count is not consumed (effective `total_CFs=1`).

2.5 Metrics and Statistical Inference

Primary metrics are fidelity, stability, sparsity, faithfulness gap, and cost (milliseconds). Aggregation is instance → run → block. EEU is not used for confirmatory inference in this draft. The multi-metric framing follows recommendations from recent XAI evaluation surveys and critical reviews (Kadir et al., 2023; Pawlicki et al., 2024; Adadi and Berrada, 2018; Arrieta et al., 2020).

Faithfulness and stability are computed as:

$$\Delta_k = \left|f(x) - f\left(x_{\text{mask-top-}k}\right)\right|, \quad \rho = \operatorname{corr}\left(|w_i|, |f(x) - f(x_{-i})|\right), \tag{1}$$

$$x^{(t)} = x + \epsilon^{(t)},\ \epsilon^{(t)} \sim \mathcal{N}(0, \sigma^2), \quad S = \frac{2}{T(T-1)} \sum_{a<b} \cos\left(e^{(a)}, e^{(b)}\right). \tag{2}$$

Inferential workflow:

1. Friedman omnibus test per metric on complete (g, n) blocks.

2. Nemenyi post-hoc localization when Friedman is significant.

3. Two-sided paired Wilcoxon SHAP–LIME tests on matched (g, s, n) cells (45-cell primary set; 65-cell sensitivity set).

Multiplicity handling uses Nemenyi within each metric and Holm-Bonferroni across the five primary metrics for each inferential family (Friedman, Wilcoxon-45, Wilcoxon-65). Reported uncertainty/effect summaries include mean, standard deviation, CV, confidence intervals (where generated), Kendall's W, Cohen's d_z, and median paired differences.

Artifact qualification is deterministic: malformed or empty runs are excluded, no synthetic reconstruction is used, and omnibus tests are restricted to block-complete contexts.

2.6 FOM-7 Origin, Operational Definition, and Validity Boundaries

FOM-7 denotes the **F**ramework **O**peration **M**ethod with seven sequential quality gates. It was derived from two inputs: (i) literature-grounded benchmark requirements for transparent, reproducible evaluation (Canha et al., 2025; Agarwal et al., 2022; Hedström et al., 2023), and (ii) failure modes observed in the EXP1 pilot/calibration stream (Table 1), especially protocol drift, artifact integrity failures, and non-equivalent analysis exports. To convert these risks into explicit controls, the benchmark workflow was restructured as the following stage-gated protocol:

1. **Protocol freeze:** lock code/configuration versions, factors, and inferential plan before confirmatory runs.

2. **Controlled batch execution:** execute runs from declared manifests with fixed seeds and logged runtime context.

3. **Artifact integrity audit:** detect and exclude malformed, empty, or schema-incompatible outputs.

4. **Harmonization to analysis-ready tables:** standardize keys and metric fields into deterministic, merge-safe inferential tables.

5. **Inferential export:** generate deterministic omnibus and paired-test artifacts from qualified inputs only.

6. **Reproducibility profiling:** quantify run-to-run dispersion and coefficient-of-variation behavior on replicated settings.

7. **Claim-ready reporting:** permit inferential claims only when all prior gates are satisfied and traceable to source artifacts.

Progression requires gate satisfaction at each stage; failed gates downgrade affected outputs to descriptive-only status.

The reproducibility package includes code, experiment configurations, model and preprocessor artifacts, per-run outputs, analysis exports, and figure scripts. Primary repository anchors used in this paper are:

- `configs/experiments/exp2_scaled/manifest.yaml` (declared EXP2 run grid),

FOM-7 flow: Freeze → Execute → Audit → Harmonize → Export → Profile → Report

Figure 2: Compact operational flow of FOM-7 from protocol lock to claim-ready reporting.

- `experiments/exp2_scaled/results/` (qualified per-run outputs),
- `outputs/paper_analysis/paper_comparison.csv` (aggregated comparison export),
- `scripts/generate_paper_a_figures.py` (figure-generation entry point).

Validity controls and residual risks are summarized as follows:

1. **Internal validity:** explicit missingness audit and block-complete inference. A standardized runtime constraint protocol (Cores=1, RAM=4GB, Timeout=300s) is formally enforced via the `ResourceGuard` utility to ensure architectural comparability; residual risk from non-random coverage gaps.
2. **Construct validity:** metric definitions are operationalized in the `src/metrics/` directory. For technical traceability, conceptual definitions reported in Section 2.5 are anchored to implementation classes: `FidelityMetric`, `StabilityMetric`, `SparsityMetric`, `FaithfulnessMetric` (gap field), and `CostMetric`.
3. **Statistical conclusion validity:** matched-design non-parametric tests with multiplicity control; residual risk from absent equivalence bands.
4. **External validity:** diversity across model families and seeds, but single tabular dataset scope.
5. **Implementation validity:** effective runtime parameter bindings are disclosed; harmonization changes may shift absolute levels.

Scope of claims: conclusions in Section 3 are comparative and bounded to this benchmark setting (Adult tabular task, declared models/explainers, and artifact-qualified EXP2 subsets).

3 Results

3.1 Overview of Evaluation Setup

This section reports results from two evidence streams: EXP2 robustness (global and paired method comparisons) and EXP1 reproducibility (variance profiling). Compared explainers are SHAP, LIME, Anchors, and DiCE. Reported metrics are Fidelity, Stability, Sparsity, Faithfulness Gap, and Cost (ms). This multi-metric framing treats explanation quality as a vector of properties rather than a single scalar score (Kadir et al., 2023; Pawlicki et al., 2024; Arrieta et al., 2020). Metric definitions and the inferential workflow are specified in Section 2.5. "15 complete model-size blocks" denotes the block-level EXP2 units used for global inference, and "45 matched configs" denotes the SHAP–LIME subset with shared $(\text{model}, \text{seed}, N)$ cells on `logreg/rf/xgb`. Metric direction is defined in Table 3, and cohort qualification counts are reported in Table 2.

3.2 Global Trade-offs Across Methods (Quality vs Cost)

Table 4: Global method means (EXP2 robustness: 15 complete model-size blocks).

Method	Fidelity	Stability	Sparsity	Faithfulness Gap	Time (ms)
SHAP	0.8176	0.7377	0.2264	0.4474	685220.23
LIME	0.5602	0.0144	0.0846	0.3342	3660.68
Anchors	0.3853	0.0006	0.0928	0.2382	25326.92
DiCE	0.1716	0.3602	0.0164	0.0988	16306.50

SHAP has the highest Fidelity (0.8176), highest Stability (0.7377), and highest Faithfulness Gap (0.4474). DiCE has the lowest Sparsity value (0.0164), while LIME has the lowest Cost (3660.68 ms). Anchors shows very low Stability (0.0006) with Cost (25326.92 ms) above LIME and DiCE.

The reported means show an explicit quality–cost frontier: SHAP occupies the strongest quality region on Fidelity/Stability/Faithfulness Gap with the highest runtime (685220.23 ms), whereas LIME occupies the lowest-cost region with lower Fidelity and Stability. DiCE provides the lowest Sparsity value with lower Fidelity and Faithfulness Gap. This trade-off pattern is consistent with mechanistic differences between local surrogate and additive-attribution families (Ribeiro et al., 2016; Lundberg and Lee, 2017).

3.3 Global Statistical Evidence Across Methods

Table 5: Friedman tests across methods.

Metric	Statistic	p-value	Significant
Fidelity	42.12	3.78e-09	Yes
Stability	43.88	1.60e-09	Yes
Sparsity	35.64	8.92e-08	Yes
Faithfulness Gap	45.00	9.25e-10	Yes
Cost	27.72	4.16e-06	Yes

For every metric, the Friedman test rejects the null of equal performance across methods. This establishes global between-method differences, but does not identify pairwise direction without post-hoc localization. Post-hoc localization uses Nemenyi pairwise comparisons within each metric, with Holm-Bonferroni adjustment across the five primary metrics. The corresponding effect sizes indicate medium-to-very-large between-method separation across all five metrics.

3.4 Targeted Pairwise Comparison (SHAP vs LIME on Shared Families)

Within this matched subset, SHAP is higher on Fidelity, Stability, and Faithfulness Gap, while LIME has lower Cost and lower Sparsity value. The reported p-values indicate non-zero paired differences for all listed metrics. Practically, the quality gains for SHAP are accompanied by a runtime penalty relative to LIME under shared cells.

Table 6: Friedman effect sizes (Kendall's $W = \chi^2/[N(k-1)]$; $N = 15$, $k = 4$).

Metric	χ^2	Kendall's W
Fidelity	42.12	0.936
Stability	43.88	0.975
Sparsity	35.64	0.792
Faithfulness Gap	45.00	1.000
Cost	27.72	0.616

Table 7: Paired Wilcoxon comparison on 45 matched configurations (`logreg/rf/xgb`).

Metric	SHAP Mean	LIME Mean	Wilcoxon p-value
Fidelity	0.8112	0.5556	5.68e-14
Stability	0.8013	0.0154	5.68e-14
Sparsity	0.3156	0.0845	5.68e-14
Faithfulness Gap	0.3924	0.3408	5.68e-14
Cost (ms)	1258.72	210.45	2.29e-06

This comparison is restricted to `logreg/rf/xgb` matched configurations and is not directly generalizable to non-matched families or to methods outside the SHAP–LIME pair, including rule-based and counterfactual families (Ribeiro et al., 2018; Mothilal et al., 2020).

3.5 Reproducibility and Variance Profile

EXP1 repeated-run evidence indicates lower dispersion for quality-oriented metrics and higher dispersion for runtime behavior. Quality metrics remain relatively stable across repeated seeds, while Cost varies more strongly for SHAP variants, consistent with recent calls for variance-aware benchmark reporting (Agarwal et al., 2022; Hedström et al., 2023).

- Metric stability (quality): Fidelity CV ≤ 0.0634, Stability CV ≤ 0.0826, Sparsity CV ≤ 0.0441, Faithfulness-gap CV ≤ 0.0358.
- Runtime instability (cost): Cost CV reaches 0.225 for SHAP variants.
- Decision implication: quality rankings are more repeatable than runtime behavior under the reported repeated-run protocol.

3.6 Synthesis and Implications for Practice

- When the objective is maximizing Fidelity/Stability/Faithfulness Gap and compute budget is permissive, SHAP is the supported choice in this benchmark.
- When compute or latency is constrained, LIME is the supported low-cost option, with lower Fidelity and Stability than SHAP.
- When conciseness (Sparsity) is the primary objective, DiCE provides the lowest reported Sparsity value, with lower Fidelity and Faithfulness Gap.

- Methods are not statistically interchangeable: Friedman tests reject the null of equal performance across methods for all five primary metrics.
- On the shared SHAP–LIME subset, quality improvements for SHAP are paired with higher runtime cost.

Interpretation limits:

- Anchors stability values near zero should be interpreted with implementation caveats: effective runtime behavior uses fixed precision threshold 0.95 and binary rule-membership importance vectors, which can reduce perturbation-based similarity under this protocol.
- Method rankings are currently validated only on one tabular benchmark (Adult); external generalization requires replication on additional datasets and modalities before cross-domain claims are made.

4 Conclusion

The reported metrics indicate a multi-objective frontier rather than a single best method: SHAP is strongest on the reported quality-oriented metrics, LIME is strongest on runtime efficiency, and DiCE is strongest on the reported sparsity criterion. Global tests reject the null of equal method performance across the primary metrics, and paired SHAP–LIME analysis indicates a quality–cost trade-off on shared model families. This reinforces multi-objective interpretation guidance in current XAI evaluation literature (Pawlicki et al., 2024; Adadi and Berrada, 2018; Arrieta et al., 2020).

Methodologically, the current paper contributes FOM-7 (Framework Operation Method, seven-stage gating) as an auditable benchmark operation protocol that connects protocol freezing, controlled execution, integrity audit, deterministic inference export, and claim-traceable reporting. This contribution is operational and reproducibility-focused rather than algorithmic.

- **Takeaway:** method choice should be objective-driven (quality, sparsity, or runtime), not based on a single aggregated score.
- **Takeaway:** statistical evidence supports method differentiation at the omnibus level; pairwise direction must be interpreted within matched-design limits.
- **Limitation:** Systematic missingness for the DiCE explainer (due to search-space timeouts) and isolated serialization corruption for SVM-SHAP qualify the confidence in full-grid frontier stability. The reporting caveats document a detailed 17-artifact diagnostic.
- **Next steps:** Maintain the versioned reproducibility bundle with immutable DOI (code, configs, run outputs, inferential exports, and figure scripts) incorporating the final 283 analyzed artifacts.
- **Next steps:** replicate this quantitative protocol on additional tabular datasets to test ranking stability under dataset shift.

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